

Life Cycle Assessment of Small-Scale Tofu Production in Semarang City: Hotspot Analysis Using CML-IA and AWARE Methods

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Abstract

The tofu industry is one of the most important food-processing sectors in Indonesia, given its high consumption rate and notable nutritional value. However, the environmental impacts of each production stage have not been comprehensively identified. Life Cycle Assessment (LCA) with a gate-to-gate system boundary was applied to evaluate the environmental impacts of tofu production at a small and medium-sized enterprise (SME) in Semarang City. The LCA was conducted using SimaPro 10.3 with the CML-IA Baseline method, covering five impact categories: Global Warming Potential (GWP), Ozone Layer Depletion (ODP), Fresh Water Aquatic Ecotoxicity (FAET), Acidification, and Eutrophication. Water use was additionally assessed using the AWARE method. The results demonstrate that the washing and boiling stages are the primary contributors to environmental impacts across all assessed categories. Normalization results indicate that Fresh Water Aquatic Ecotoxicity is the most dominant impact category (1.33×10^3 kg 1,4-DB eq), followed by Global Warming Potential (5.28×10^3 kg CO₂ eq). The principal environmental hotspots are the organic wastewater generated during the washing, soaking, and coagulation stages, as well as firewood combustion during boiling. Recommended improvement strategies include the implementation of wastewater treatment systems, water recycling practices, and fuel substitution.

Keywords: Life Cycle Assessment; tofu production; CML-IA Baseline; AWARE; environmental hotspot; Semarang

1. Introduction

Industrial activity is one of the primary drivers of global economic growth [1]. The tofu industry represents a significant food-processing sector in Indonesia owing to its high consumption rate and notable nutritional value. According to Statistics Indonesia (BPS), average per capita tofu consumption reached approximately 0.163 kg per week in 2024, indicating that tofu remains a staple food in many Indonesian households [2]. This high demand has driven increased production activity, resulting in greater resource and energy consumption, as well as the generation of waste and emissions that may cause environmental pollution if not properly managed [3].

Such impacts are particularly evident in river water quality, as the discharge of industrial wastewater can cause Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD) levels to exceed regulatory standards. In the case of the tofu industry in Mojokerto, liquid waste was reported to contain BOD levels exceeding 2,000 mg/L and COD levels exceeding 5,000 mg/L, far above the standard thresholds of approximately 50 to 100 mg/L [4]. Furthermore, the environmental impacts associated with each stage of tofu production have not yet been

comprehensively identified. A systematic approach such as Life Cycle Assessment (LCA) is therefore required to evaluate the full environmental burden of the tofu production process [5].

Life Cycle Assessment (LCA) is a standardized method used to systematically evaluate the environmental impacts of a product or process based on all input and output flows throughout its life cycle [6]. This method is capable of identifying production stages that constitute environmental hotspots, thereby serving as the scientific basis for developing targeted environmental improvement strategies [7]. Previous studies have applied LCA to assess the environmental impacts of the tofu industry; however, these studies predominantly focused on climate change impact categories and thus failed to comprehensively evaluate other environmental concerns, such as water pollution loads, water scarcity, and solid waste generation across all production stages [8, 9].

Therefore, this study aims to analyze the environmental impacts of tofu production in Semarang City using an LCA approach with a gate-to-gate system boundary, encompassing all stages from washing to packaging, in order to identify environmental hotspots and formulate more sustainable improvement strategies.

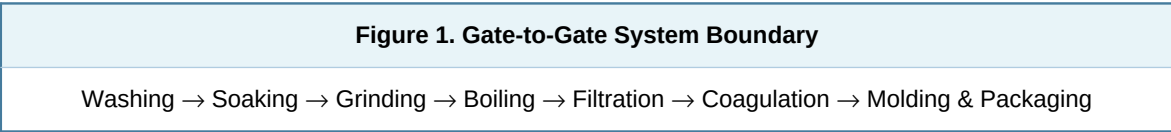
2. Methods

This research utilized a quantitative methodology based on Life Cycle Assessment (LCA) to assess the environmental impacts of tofu production at an SME in Semarang City. The evaluation was performed in compliance with **SNI ISO 14040:2016** and **SNI ISO 14044:2016**, comprising four sequential phases: (1) goal and scope definition, (2) life cycle inventory analysis, (3) life cycle impact assessment, and (4) interpretation. Primary data were obtained through direct field measurements, structured interviews with the facility operator, and on-site documentation during a complete production batch. Secondary data were sourced from peer-reviewed scientific literature and the Ecoinvent v3 database within SimaPro 10.3 [6].

2.1 Goal and Scope Definition

This study aims to assess the environmental impacts of tofu production at the facility level, focusing on water use, energy consumption, and waste generation at each production stage. The **functional unit** was established as **1 kg of tofu produced**, consistent with functional units used in comparable LCA studies of tofu and soy-based food processing. All inventory data were initially collected on a per-batch basis representing one complete production day, with 1,400 kg of soybeans as raw material input yielding 3,681 kg of tofu.

The **system boundary** adopts a gate-to-gate approach, encompassing all unit processes from the initial washing of soybeans to final molding and packaging (Figure 1). Upstream operations, including soybean farming and transportation, as well as downstream activities such as distribution, consumption, and end-of-life management, were excluded from the system boundary. This delineation is consistent with prior gate-to-gate LCA investigations of similar Indonesian tofu SMEs [8, 9].



2.2 Life Cycle Inventory (LCI) Analysis

The inventory analysis phase involves the identification and collection of quantitative data on all input and output flows for each unit process within the product system, in accordance with the defined functional unit [11]. Data collected include soybean inputs, water and energy consumption, and waste outputs comprising liquid waste, solid waste, and atmospheric emissions. The complete inventory dataset is presented in Table 1.

Table 1. Life Cycle Inventory Data of Tofu Production (per batch / production day).

Stage	Input	Qty	Unit	Output	Qty	Unit
Washing	Soybeans	1,400	kg	Washed soybeans	1,540	kg
	Water	2,800	L	Wastewater	2,660	L
	Electricity	0.60	kWh	CO ₂ emissions	0.51	kg
Soaking	Soybeans	1,540	kg	Hydrated soybeans	3,640	kg
	Water	2,700	L	Wastewater	600	L
	Electricity	0.45	kWh	CO ₂ emissions	0.38	kg
Grinding	Hyd. soybeans	3,640	kg	Soybean slurry	7,840	kg
	Water	4,200	L	Wastewater	0	L
	Electricity	17.00	kWh	CO ₂ emissions	14.45	kg
Boiling	Soybean slurry	7,840	kg	Boiled slurry	15,652	kg
	Water	8,400	L	Steam loss	588	L
	Firewood	800	kg	CO ₂ emissions	1,397	kg
				CH ₄ emissions	0.0499	kg
				N ₂ O emissions	0.3744	kg
Filtration	Boiled slurry	15,652	kg	Soy milk	14,052	kg
				Okara (residue)	1,600	kg
Coagulation	Soy milk	14,052	kg	Tofu curd	3,681	kg
	Coagulant water	421	L	Whey	10,434	L
Molding & Pkg	Tofu curd	3,681	kg	Final product	3,681	kg
	Water	100	L	Wastewater	100	L

2.3 Life Cycle Impact Assessment (LCIA)

The LCIA phase evaluates and quantifies the potential environmental impacts generated by the product system based on the compiled inventory data [6]. In this study, the **CML-IA Baseline** method was applied using SimaPro 10.3, covering five impact categories: Global Warming Potential (GWP₁₀₀, kg CO₂ eq), Ozone Layer Depletion (ODP, kg CFC-11 eq), Fresh Water Aquatic Ecotoxicity (FAET, kg 1,4-DB eq), Acidification (AP, kg SO₂ eq), and Eutrophication (EP, kg PO₄³⁻ eq). Water use and potential water scarcity were additionally assessed using the **AWARE (Available WATER REMaining)** method to evaluate water deprivation relative to regional water availability.

2.4 Interpretation

The interpretation phase contextualizes LCIA results to identify hotspot impact categories and the production stages contributing the greatest environmental burdens. Normalization was subsequently applied using the CML-IA Baseline reference values to compare impact categories expressed in different units on a common dimensionless scale, thereby enabling the relative significance of each impact to be determined. Interpretation was focused strictly within the gate-to-gate system boundary to ensure that results accurately reflect the stages exerting the greatest environmental influence [6].

3. Results and Discussion

3.1 Environmental Impact of Tofu Production

The LCIA reveals that tofu production generates measurable environmental impacts across all five assessed categories. The quantitative results per production stage are summarized in Table 2.

Table 2. Environmental Impact of Tofu Production per Batch by Production Stage.

Impact Category	Unit	Total	Washing	Soaking	Grinding	Boiling	Coagulation	Molding
GWP	kg CO ₂ eq	5.28×10 ³	3.56×10 ³	0.945	35.8	1.63×10 ³	55.8	5.9
ODP	kg CFC-11 eq	6.11×10 ⁻³	5.93×10 ⁻³	2.1×10 ⁻⁵	7.93×10 ⁻⁵	1.07×10 ⁻⁴	5.02×10 ⁻⁵	1.29×10 ⁻⁵
FAET	kg 1,4-DB eq	1.33×10 ³	1.22×10 ³	0.651	24.6	70.6	14.1	2.75
Acidification	kg SO ₂ eq	4.59	3.53	0.00245	0.0925	0.723	0.226	0.0176
Eutrophication	kg PO ₄ ³⁻ eq	9.10	8.30	0.0034	0.128	0.409	0.250	0.00838

Note: The filtration stage does not appear in the impact results because no direct energy inputs or emissions were entered in the inventory model for this stage; it therefore generates no characterization values in SimaPro.

3.1.1 Global Warming Potential (GWP)

Total GWP amounts to 5.28×10³ kg CO₂ eq. The washing stage is the largest contributor at 3.56×10³ kg CO₂ eq (approximately 67.4%), followed by the boiling stage at 1.63×10³ kg CO₂ eq (approximately 30.9%). The dominance of the washing stage is attributed to the large volumes of water consumed during soybean cleaning, which indirectly require energy for water supply and distribution, thereby generating greenhouse gas emissions. In addition, organic-laden washing wastewater has the potential to generate methane (CH₄) if discharged without treatment. The boiling stage contributes significantly to GWP owing to the combustion of firewood, which releases CO₂, CH₄, and N₂O, all of which are potent greenhouse gases with long-term atmospheric warming effects.

3.1.2 Ozone Layer Depletion (ODP)

Total ODP registers 6.11×10⁻³ kg CFC-11 eq, with the washing stage dominating at 5.93×10⁻³ kg CFC-11 eq (approximately 97.1%). This impact is not attributed to water consumption per se, but rather to the energy-intensive infrastructure required for water supply and distribution within the production chain, which generates indirect emissions linked to ozone-depleting substances. Although this category contributes the least in normalized terms, its

disproportionate concentration in the washing stage highlights the indirect environmental burden of high water use.

3.1.3 Fresh Water Aquatic Ecotoxicity (FAET)

Total FAET reaches 1.33×10^3 kg 1,4-DB eq, with the washing stage as the primary contributor at 1.22×10^3 kg 1,4-DB eq (approximately 91.7%). The large volumes of organic-laden wastewater generated during soybean washing contain residual organic matter, dissolved solids, and impurities that constitute a significant toxicity risk to freshwater organisms if discharged without adequate treatment. The boiling stage contributes 70.6 kg 1,4-DB eq (approximately 5.3%) through the atmospheric deposition of combustion byproducts, including nitrogen oxides (NO_x) and sulfur oxides (SO_x), into aquatic environments.

3.1.4 Acidification

Total Acidification impact is 4.59 kg SO₂ eq. The washing stage accounts for 3.53 kg SO₂ eq (approximately 76.9%), associated with SO₂ and NO_x emissions from energy generation required for water provision. The boiling stage contributes 0.723 kg SO₂ eq (approximately 15.8%) through firewood combustion, which releases acidifying gaseous precursors that may react with atmospheric moisture to form acid deposition, potentially degrading both soil and water quality in the surrounding environment.

3.1.5 Eutrophication

Total Eutrophication impact is 9.10 kg PO₄³⁻ eq, with the washing stage dominant at 8.30 kg PO₄³⁻ eq (approximately 91.2%). This impact is driven primarily by nutrient-rich washing wastewater that contains nitrogen and phosphorus compounds derived from soybean residues. When discharged into receiving water bodies without treatment, these compounds can trigger excessive proliferation of algae and aquatic microorganisms, leading to depletion of dissolved oxygen and disruption of aquatic ecosystem function. The boiling stage contributes 0.409 kg PO₄³⁻ eq (approximately 4.5%) through atmospheric deposition of NO_x emissions generated during firewood combustion.

3.2 Water Use Assessment (AWARE Method)

Water use was assessed using the AWARE method to quantify potential water scarcity, expressed as the volume of water deprived from downstream users per unit of water consumed at a given location. The total water scarcity impact for the entire production system amounts to **515 m³ world eq** (Table 3).

Table 3. Water Use Assessment Results by Production Stage (AWARE Method).

Production Stage	Water Scarcity (m ³ world eq)	Share (%)
Boiling	294	57.09
Grinding	101	19.61
Soaking	39.3	7.63
Coagulation	38.2	7.42
Washing	35.8	6.95
Molding & Packaging	7.63	1.48
Total	515	100

The boiling stage is the dominant contributor (approximately 57.1%), reflecting both direct water consumption during soybean slurry cooking and the substantial upstream water demand embedded in the firewood supply chain. Within the AWARE framework, the environmental burden is determined not solely by the volume of water consumed, but also by the water intensity of the upstream energy system and associated supply chains. The grinding stage contributes the second largest share (approximately 19.6%), as continuous water addition is required throughout this process to achieve the appropriate slurry consistency and extraction efficiency.

3.3 Normalization

Normalization was performed using the CML-IA Baseline reference values to place all impact categories on a common dimensionless scale, enabling direct and meaningful comparison of relative environmental significance across categories with different units (Table 4).

Table 4. Normalization Results of Tofu Production Environmental Impacts (CML-IA Baseline).

Impact Category	Normalized Value	Rank
Fresh Water Aquatic Ecotoxicity (FAET)	2.57×10^{-1}	1
Global Warming Potential (GWP)	1.05×10^{-1}	2
Eutrophication	6.89×10^{-2}	3
Acidification	1.63×10^{-2}	4
Ozone Layer Depletion (ODP)	6.84×10^{-4}	5

Normalization confirms that **Fresh Water Aquatic Ecotoxicity (FAET) is the dominant environmental impact category** in this production system, indicating that tofu production exerts the greatest relative pressure on freshwater ecosystems. This finding underscores that organic liquid waste management, particularly the treatment of effluents from the washing, soaking, and coagulation stages, constitutes the most critical environmental challenge in the analyzed production system.

Although Global Warming Potential frequently receives primary attention in environmental sustainability discourse, its normalized contribution ranks second in this study, indicating that freshwater pollution represents a more acute environmental concern at the facility scale assessed. Eutrophication, which ranks third in normalized impact, further reinforces the urgency of addressing nutrient-laden wastewater streams as a priority intervention. Acidification and Ozone Layer Depletion, while present, contribute relatively minor normalized values, suggesting that their management, though important, does not constitute the primary environmental priority for this production system.

3.4 Hotspot Identification and Improvement Strategies

The integrated analysis of LCIA results and normalization outcomes identifies two primary environmental hotspots in the tofu production system, namely the generation of organic liquid waste across the washing, soaking, and coagulation stages, and the combustion of firewood during the boiling stage. These hotspots correspond, respectively, to the dominant drivers of Fresh Water Aquatic Ecotoxicity and Eutrophication (ranked first and third in normalized impact), and of Global Warming Potential and water scarcity.

With respect to the first hotspot, the organic-laden wastewater streams generated during washing, soaking, and coagulation are characterized by high BOD, COD, and nutrient loads [13,

14], and their uncontrolled discharge poses significant risks to receiving aquatic ecosystems. The implementation of anaerobic treatment systems, such as biodigesters or integrated biogas wastewater treatment units, represents an effective and technically appropriate intervention for SME-scale operations, as such systems simultaneously reduce organic loading while generating biogas as a recoverable energy resource. Complementary measures include the application of constructed wetland or biofilter systems for tertiary nutrient removal prior to effluent discharge, as well as counter-current water recycling and recirculation within the washing and soaking stages to reduce both volumetric water consumption and wastewater generation. Furthermore, the valorization of whey, for instance as a protein-rich substrate for animal feed or as a feedstock for fermentation processes, offers a viable pathway for diverting high-nutrient liquid streams away from the wastewater treatment system, thereby reducing the overall nutrient load requiring treatment [10].

With respect to the second hotspot, the boiling stage contributes disproportionately to GWP and water scarcity owing to its reliance on firewood as the primary thermal energy source. The combustion of 800 kg of firewood per production batch releases 1,397 kg CO₂, 0.0499 kg CH₄, and 0.3744 kg N₂O, collectively representing a substantial greenhouse gas burden. Transitioning to cleaner energy sources, such as liquefied petroleum gas (LPG) or biomethane derived from on-site biodigesters, would directly reduce these emissions while also decreasing the upstream water footprint associated with firewood supply chains. In parallel, the adoption of improved combustion systems or thermally insulated boiling tanks would reduce specific fuel consumption per unit of output through minimization of heat loss, while waste heat recovery from boiler exhaust could be directed toward pre-heating process water at upstream stages, thereby improving overall energy efficiency. The LCA-based analysis demonstrates that improvement interventions targeting liquid waste management and energy use efficiency at the source generate substantially greater environmental benefits than conventional end-of-pipe approaches [10, 15].

4. Conclusion

This study applied Life Cycle Assessment with a gate-to-gate system boundary to evaluate the environmental performance of small-scale tofu production at an SME in Semarang City. The assessment encompassed five impact categories under the CML-IA Baseline method, supplemented by water scarcity analysis using the AWARE method, and yielded quantitative evidence on the distribution of environmental burdens across all production stages.

The results demonstrate that tofu production generates measurable impacts across all assessed categories, with total values of 5.28×10^3 kg CO₂ eq for GWP, 6.11×10^{-4} kg CFC-11 eq for ODP, 1.33×10^3 kg 1,4-DB eq for FAET, 4.59 kg SO₂ eq for Acidification, and 9.10 kg PO₄³⁻ eq for Eutrophication. The washing and boiling stages consistently emerge as the principal contributors across all impact categories, driven by high water consumption and firewood combustion, respectively. Normalization of these results reveals that Fresh Water Aquatic Ecotoxicity constitutes the dominant environmental impact category, with a normalized value of 2.57×10^{-4} , confirming that the organic liquid waste streams generated throughout the production process represent the most critical environmental hotspot. Water scarcity, as assessed by the AWARE method, totals 515 m³ world eq, with the boiling stage accounting for 57.1% of this burden, a value that reflects not only direct water use but also the embedded water

demand of firewood procurement and combustion.

These findings indicate that environmental improvement efforts should be strategically directed toward two priority areas. The first is the establishment of structured liquid waste management systems, including anaerobic treatment, nutrient recovery, and water recycling, to address the dominant freshwater ecotoxicity and eutrophication burdens arising from washing, soaking, and coagulation effluents. The second is the substitution of firewood with cleaner fuel alternatives and the adoption of energy-efficient thermal systems in the boiling stage, which would simultaneously reduce greenhouse gas emissions, acidification potential, and water scarcity impacts. These conclusions are consistent with broader findings in the LCA literature on soy-based food processing, which collectively emphasize the critical importance of wastewater management and energy source selection as the primary levers for environmental performance improvement in tofu production systems [8, 9]. The findings of this study provide actionable, evidence-based guidance for plant operators and policymakers seeking to improve the environmental sustainability of SME-scale tofu production, and may inform the development of sector-specific environmental management standards for the tofu industry in Semarang City and Central Java.

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★ = reference validity could not be independently confirmed; authors should verify prior to journal submission.

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